Mohamed Osman Mohamed

Senior Backend Engineer

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LeetCode My Portfolio

Profile

Senior backend engineer who designed and built ABS Courier's logistics platform from scratch, replacing a purchased legacy system. It now runs 20M+ shipments a year for 1,000+ users. Strongest in TypeScript, Node.js and NestJS, PostgreSQL performance (partitioning, read replicas, query tuning), queue-based processing with Bull and Redis caching. Led a team of 5 backend engineers, and also ships React, Next.js and React Native frontends.

Professional Experience

ABS Courier

2024/11 – 2026/10
Cairo, Egypt

Backend Team Leader

- Designed and built the company's logistics platform from scratch, replacing a purchased legacy system; it now handles 20M+ shipments per year for 1,000+ users.
- Gathered requirements directly from operations, branch and fleet management teams and translated them into system design.
- Partitioned high-volume shipment tables by date, keeping query times flat as the dataset grew.
- Introduced PostgreSQL read replicas behind PgBouncer, routing lag-tolerant reporting reads off the primary.
- Ran zero-downtime migrations on live high-volume tables.
- Moved bulk ingest, WhatsApp notifications and tracking events onto Bull queues processed by a dedicated worker.
- Containerised the stack with Docker and introduced a Redis caching layer, improving response times ~3.7x.
- Built the public shipment-tracking and partner-facing APIs.
- Started the platform's re-architecture into a multi-tenant SaaS for external courier companies.
- Led a team of 5 backend engineers: review standards, onboarding and mentoring.

2024/05 – 2024/11
Cairo, Egypt

Full Stack Developer

- Joined while the company was running a purchased legacy system; proposed NestJS and PostgreSQL as the foundation for a purpose-built replacement and owned the architecture decisions behind it.
- Took ownership of the backend within six months and moved into the team lead role.

2024/01 – 2026/09
Giza, Egypt

Digital Egypt Cubs Initiative "DECI"
Instructor

- Teach coding and robotics to young learners as part of the Egypt Digital Cubs Initiative

2023/06 – 2024/05
Al Rehab City, Egypt

NasNav
Software Developer

- Built web applications with React.js and cross-platform mobile apps with React Native, integrating with backend APIs.

2022/06 – 2023/06
El Sheikh Zayed, Egypt

Teqneia ICT
Software Developer

- Built responsive React.js web applications for client projects.

Education

2018 – Present
Giza, Egypt

Faculty of Science, Cairo University
Bachelor of Computer Science

Skills

Backend & Architecture: Node.js (NestJS, Express) | TypeScript | REST & GraphQL APIs | Event-driven design | Bull queues & background workers | Multi-tenant architecture

Databases: PostgreSQL (partitioning, replication, indexing, query tuning) | PgBouncer | Redis (caching, queues) | MySQL | MongoDB

Infrastructure & Observability: Docker | AWS (EC2, S3, RDS, Aurora) | Nginx | CI/CD | OpenTelemetry | Prometheus | Loki | Git

Frontend: React.js | Next.js | React Native | TypeScript

Practices: Architecture Decision Records | Zero-downtime migrations | Code review | Technical mentoring

Languages

Arabic — Fluent

English — Proficient

Awards

ECPC
2022/08/15
Egyptian Collegiate Programming Contest — competitive algorithmic problem solving.